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Decentralized Certification on Bitcoin

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Introduction

- Verification of a document is essential for maintaining organizational trust
- The centralization of verification incurs significant expenses
- Bitcoin is the first decentralized network that is: permissionless, highly tolerant to DDoS attacks, and based on Proof-of-Work (PoW)

Motivation

- Legacy Bitcoin solutions are not scalable
- **Using alternative networks such as Ethereum introduces different security and trust considerations**, increasing consensus complexity and **censorship risks**.
- Major Bitcoin upgrades (SegWit and Taproot) enable new possibilities on the base layer (**L1**) by improving **script efficiency** and **privacy**, without relying on external layers.

1. Foundation: Malleability and Legacy Anchoring Techniques

- **Transaction Evolution (Legacy):** Initial transactions (P2PKH, P2SH) suffered from **malleability**, a vulnerability allowing **txid** modification by altering the **ECDSA signature encoding** without invalidating the signature.
- **SegWit (2017):** Implemented **BIP141** to resolve malleability by moving signatures to the separate **Witness** section, introducing new types (P2WPKH, P2WSH).
- **Data Anchoring Limitations:** Initial methods lacked standardized space for arbitrary data:
  - **P2PKH/Fake Addresses:** Encoded data inside unspendable addresses, resulting in **high cost** (burned Bitcoin) and **limited capacity** (20 bytes/output).

2. The OP_RETURN Standard and Identified Problems

- **OP_RETURN Dominance:** Established as the **standard and dominant method** for embedding non-monetary data in Bitcoin, utilized heavily by services like **Omni** and **Veriblock** Strehle and Steinmetz, "Dominating OP Returns: The Impact of Omni and Veriblock on Bitcoin".
- **OP_RETURN Mechanism:** Uses an unspendable transaction output (**UTXO**) limited to **80 bytes**. It is effective for minimizing **UTXO bloat** but contributes to overall blockchain growth ($\approx 10\ GB$ in the study period).
- **Identified Design Flaws for Verification:** Legacy anchoring methods present two critical flaws for advanced verification protocols:
  1. **Lack of Privacy: OP_RETURN content is fully public**, exposing the anchored information.

3. Alternative Systems and the Revocation Dilemma

- **Ethereum (EVM/Smart Contracts):** Alternative networks use the $\mathbf{EVM}$ ($\mathbf{Turing-complete}$) model to manage $\mathbf{complex\ certification\ logic}$ (identity, permissions, state changes) Wang et al., "Blockchain-Based Certificate Transparency and Revocation Transparency"; Rustemi et al., "A Systematic Literature Review on Blockchain-Based Systems for Academic Certificate Verification".
- **Trade-off:** While flexible, these solutions depend on $\mathbf{costly\ execution}$ ($\mathbf{Gas\ Fees}$) and programmable complexity ($\mathbf{Solidity}$, $\mathbf{EVM}$), introducing new trust assumptions and $\mathbf{consensus\ complexity}$.
- **Institutional Challenges:** Studies on academic certification fraud ($\$\mathbf{2\ billion\ market}$) highlight that immutability is a challenge:

4. Literature Summary and Research Gap

- **Literature Summary:** The field presents a clear trade-off:
  - **Legacy Bitcoin (OP_RETURN)**: High **immutability** but **low scalability/privacy** for verification logic.
  - **Smart Contracts (EVM)**: High **scripting flexibility** but **high operational cost** and **lower confidence in base layer immutability**.
- **Research Gap:** The literature **has not yet investigated** how **Taproot** (activated 2021) specifically addresses this trade-off for decentralized certificate verification.
- **Taproot's Unique Solution (Justification):**
  - **MAST (Merkelized Abstract Syntax Trees)**: Allows encoding complex verification logic (**revocation**) with a minimal **transaction footprint**.
  - **Schnorr Signatures**: Provides **efficiency and key aggregation** for the protocol's

1. Research Questions

- How can the tools applied by **Taproot** be leveraged to deploy a decentralized verification protocol?
- To what extent does using **MAST** improve **privacy**, **scalability**, and **spatial/computational complexity**?
- What is an appropriate methodology to add signature **key rotation** and **certificate revocation** to the protocol?

2. Specific Objectives

- Design the **Merkle structure of scripts** for verifying the integrity of issued documents.
- Design software components for **ownership verification** (digital signature) and **document signature key rotation**.
- Design a software component that tracks **revocation** to verify the validity of issued documents.
- Deploy a **Proof of Concept (PoC)** on a Bitcoin test network.
- **Measure** the **scalability**, **time/space complexity**, and network **costs (vBytes)** of the proposed protocol.

Protocol Structure and Verification Flow

Protocol Overview (Three Phases)

- **Phase I: Design** ($\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d}$ Objectives) $\rightarrow$ Protocol architecture based on Taproot/MAST.
- **Phase II: Prototyping** $\rightarrow$ $\mathbf{PoC}$ deployment on Testnet/Signet using Python.
- **Phase III: Validation** $\rightarrow$ Experimental analysis of $\mathbf{Scalability}$ and $\mathbf{vByte\ Cost}$.

Methodology

Core Roles and Components

- **Issuer (Organization):** Provides the document and a $'\mathbf{Proof\ of\ Validation}'$ file to the user.
- **Verifier (User):** Requires only the document, the proof, and access to the $\mathbf{public\ Bitcoin\ history}$.
- **Full Verification** requires three successful checks: $\mathbf{Integrity}$, $\mathbf{Ownership}$, and $\mathbf{Revocation}$ status.

Note: All computational procedures (hashing, Merkle path) are executed $\mathbf{off-chain}$.

Key Conclusions and Thesis Contributions

- **Decentralization Advantage:** Demonstrated that Bitcoin's inherent **availability**, **DoS resistance**, and **immutability** surpass the limitations of **centralized PKI**.
- **Taproot's Critical Role:** Identified the gap in the literature regarding document verification using **Taproot (2021)**. Taproot effectively **overcomes L1 limitations** regarding **scalability** and **scripting privacy** faced by legacy solutions (OP_RETURN, OpenTimestamps).

Protocol Contributions: The proposed methodology leverages Taproot's cryptographic tools to enable:

- **Integrity Verification** via **MAST anchored hashes**.
- **Ownership Verification** via **Schnorr Digital Signatures**.
- **Document Revocation** and **Key Rotation** via **chained transactions** (Key-Path spending).
- **Privacy Preservation** by only revealing the necessary verification path.

Future Work and Limitations

- **Scope Limitations:** The protocol does not address $\mathbf{content\ validity}$ or $\mathbf{long-term\ preservation}$ of documents (user responsibility).
- **Future Directions:** Study the implementation of $\mathbf{penalties\ and\ incentives}$ for invalid proof presentation, and explore $\mathbf{long-term\ preservation\ mechanisms}$.

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